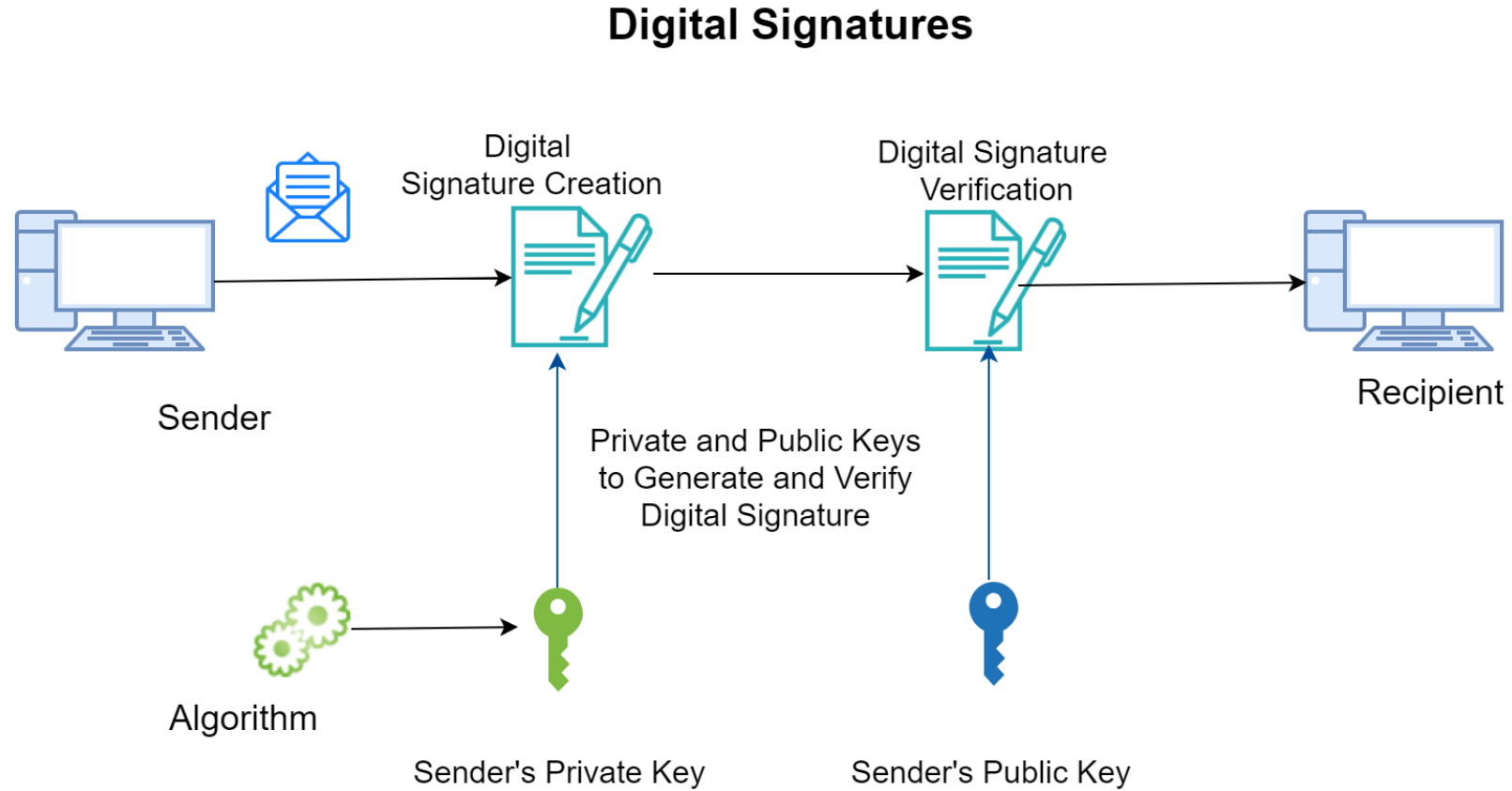


Digital signature



Digital signature

- Algorithms:

- sk : private key, pk : public key, m : message, σ : signature
- $(pk, sk) = \text{Gen}(1^n)$
- $\sigma = \text{sign}_{sk}(m)$
- $\{0,1\} \leftarrow \text{Verify}_{pk}(\sigma, m)$

Correctness: $\text{Verify}_{pk}(\sigma, m) = 1$ for $(pk, sk) \leftarrow \text{Gen}(1^n)$, any message $m \in \mathcal{M}$, $\sigma = \text{sign}_{sk}(m)$

Security

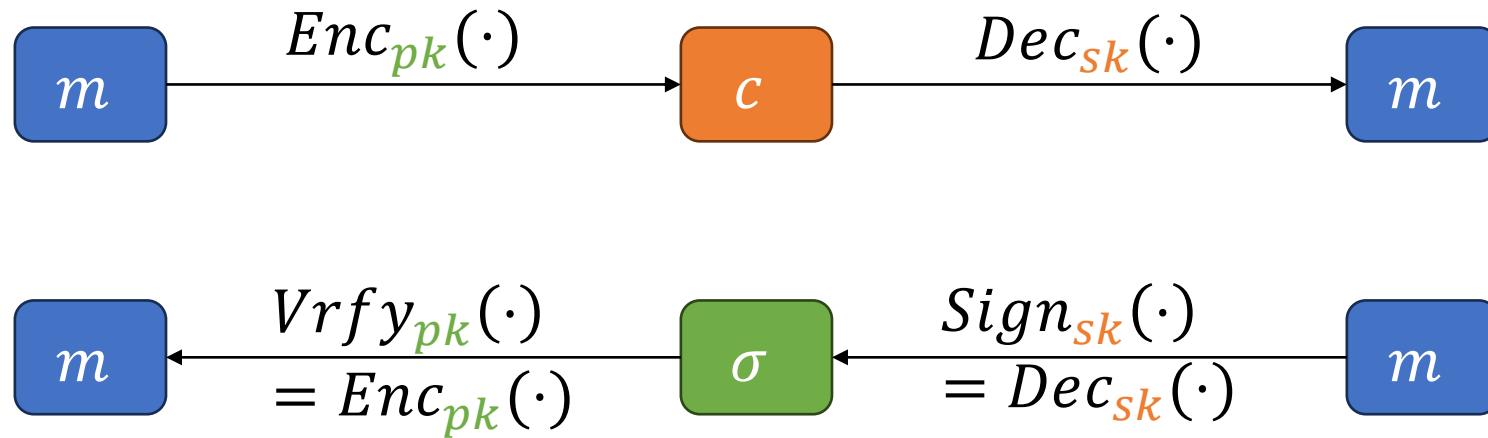
Experiment $\text{Sig_forge}_{A,\Pi}(n)$:

1. (pk, sk) are generated using $\text{Gen}(1^n)$
2. Adversary A is given pk , and **oracle access to $\text{sign}_{sk}(\cdot)$** . Let Q be the set of all queries made by A
3. A outputs (m, σ)
4. Experiment outputs 1 if (1) $\text{verify}_{pk}(m, \sigma) = 1$ and (2) $m \notin Q$

Definition: a signature scheme $\Pi = (\text{Gen}, \text{Sign}, \text{Verify})$ is **unforgeable/secure**, if for all PPT adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{Sig_forge}_{A,\Pi}(n) = 1] \leq \text{negl}(n)$$

Is this intuition correct?



No! Usually not correct or not secure!

Plain RSA Encryption

1. Choose two large prime numbers p and q such that $p \neq q$, randomly and independently of each other. Set $N = pq$.
2. Pick integer e coprime with $\phi(N) = (p - 1)(q - 1)$ i.e. $\gcd(e, (p - 1)(q - 1)) = 1$
3. Compute d such that

$$e \cdot d = 1 \bmod (p - 1)(q - 1)$$

Public key $pk = (N, e)$

Private key $sk = d$

• Encryption: $c = m^e \bmod N$

• Decryption: $m = c^d \bmod N$

Plain RSA signature

- Gen: choose two large prime numbers p and q such that $p \neq q$. Set $N = pq$. Pick integer e coprime with $\phi(N) = (p - 1)(q - 1)$, compute d such that $e \cdot d = 1 \bmod \phi(N)$

Public key $pk = (N, e)$ Private key $sk = d$

- $\text{sign}_{sk}(m)$: $\sigma = m^d \bmod N$
- $\text{Verify}_{pk}(\sigma, m)$: if $\sigma^e = m \bmod N$, output 1; otherwise, output 0

Correctness

- $ed \bmod \phi(N) = 1$

- $N = pq$

- $$\begin{aligned}\sigma^e &= (m^d \bmod N)^e \bmod N \\ &= m^{ed} \bmod N \\ &= m^{ed \bmod \phi(N)} \bmod N \\ &= m \bmod N\end{aligned}$$



Euler's theorem

Security Assumption

Factoring assumption:

given $N = pq$ where p and q are large prime numbers, it is computationally hard to find p and q .

RSA assumption:

1. Run $\text{Gen}(1^n)$ to obtain (N, e, d)
2. Choose a random $y \in \mathbb{Z}_N^*$
3. A is given (N, e, y) and outputs $x \in \mathbb{Z}_N^*$
4. Output 1 if $x^e = y \bmod N$

$$\Pr[\text{RSA}_A(n) = 1] \leq \text{negl}(n)$$

• Gen:

- $N = pq$, with p and q primes
- e relatively prime to $\phi(N) = (p - 1)(q - 1)$
- d inverse of $e \bmod \phi(N)$
- $pk = (N, e)$
- $sk = d$

• Sign:

- Plaintext m in $\mathbb{Z}_N^*$
- $\sigma = m^d \bmod N$

• Verify:

- $m \stackrel{?}{=} \sigma^e \bmod N$

Is plain RSA signature secure?

- **Attack 1:**

- Given m , we can't figure out $\sigma = m^d \pmod{N}$ because we don't know d .
- However, we know e !
- Just **sample a σ** and generate **$m = \sigma^e \pmod{N}$** and output **(m, σ)** .
- **(No-message attack)**

- **Attack 2:**

- For known $(m_0, \sigma_0), (m_1, \sigma_1)$, where $\sigma_0 = m_0^d$ and $\sigma_1 = m_1^d$.
- For arbitrary m , we can **query m_0, m_1 s.t. $m = m_0 m_1$** and **$\sigma = \sigma_0 \sigma_1$** .
- **(Forge a signature on an arbitrary message)**

RSA full-domain hash

- Gen: choose two large prime numbers p and q such that $p \neq q$. Set $N = pq$. Pick integer e coprime with $\phi(N) = (p - 1)(q - 1)$, compute d such that $e \cdot d = 1 \bmod \phi(N)$. **Specify a hash function $H: \{0, 1\}^* \rightarrow \mathbb{Z}_N^*$**

Public key $pk = (N, e)$ Private key $sk = d$

- $\text{sign}_{sk}(m)$: $\sigma = (H(m))^d \bmod N$
- $\text{Verify}_{pk}(\sigma, m)$: if $\sigma^e = H(m) \bmod N$, output 1; otherwise, output 0

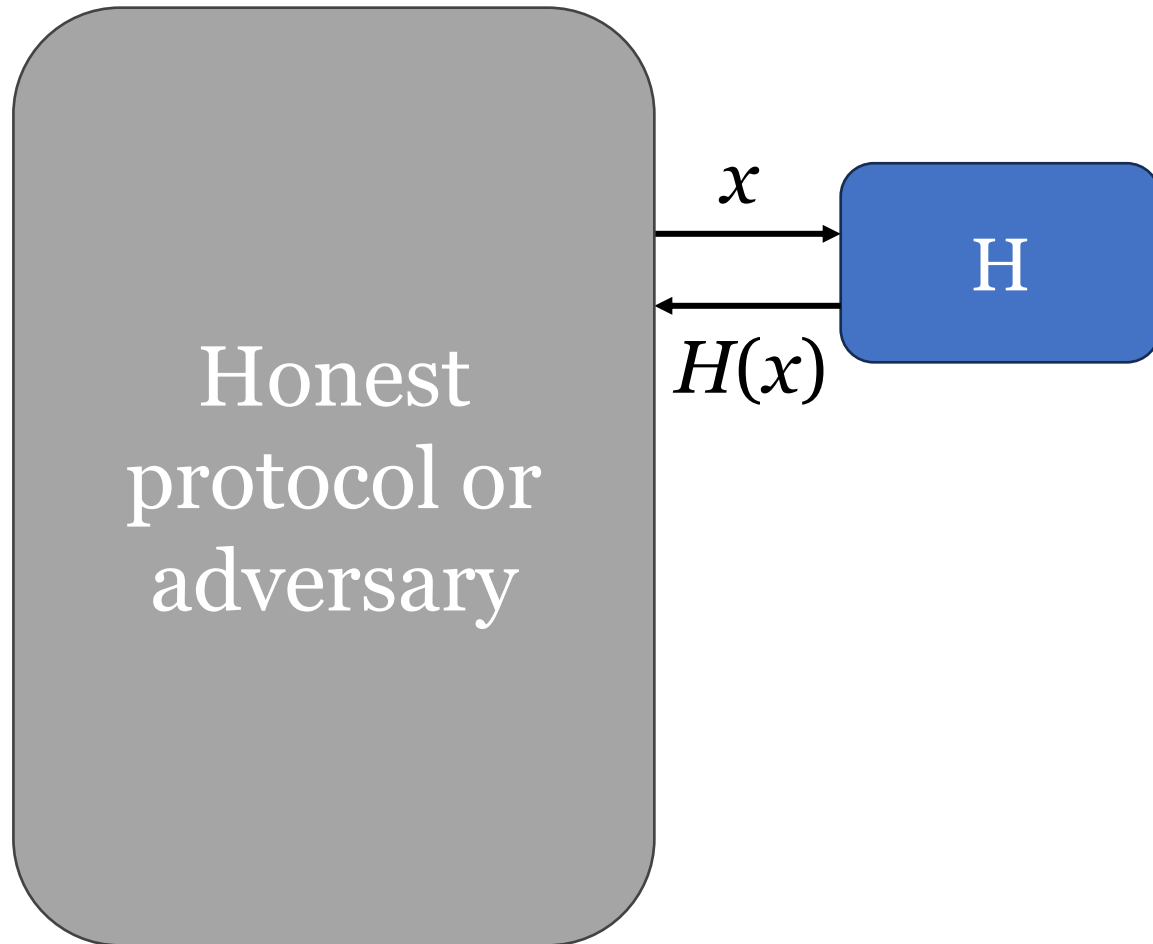
Intuition of security

- Just sample a σ , can't find $m = H^{-1}(\sigma^e \pmod N)$. **one-way**
- Can't find m, m_1, m_2 such that $H(m) = H(m_1) \cdot H(m_2)$ **one-way**
- Can't find m_1, m_2 such that $H(m_1) = H(m_2)$ **collision resistant**

No known way to choose H s.t. RSA-FDH is proven secure.

Proven secure under a random oracle!

Random Oracle Model



- If x has not been queried, $H(x)$ is uniform
- If x has been queried, return the same $H(x)$
- Record every query x in the oracle

Security

Experiment $\text{Sig_forge}_{A,\Pi}(n)$:

1. (pk, sk) are generated using $\text{Gen}(1^n)$, $H: \{0,1\}^* \rightarrow Z_N^*$ is chosen.
2. Adversary A is given pk , **oracle access to $H(\cdot)$ and $\text{sign}_{sk}(\cdot)$ such that $\sigma = H(m)^d \pmod N$** . Let **M and Q** be the set of all queries to **$H(\cdot)$ and $\text{sign}_{sk}(\cdot)$** made by A .
3. A outputs (m, σ)
4. Experiment outputs 1 if (1) $H(m) = \sigma^e \pmod N$ and (2) $m \notin Q$

Definition: a signature scheme $\Pi = (\text{Gen}, \text{Sign}, \text{Verify})$ is **unforgeable/secure**, if for all PPT adversaries A , there is a negligible function negl such that for all n ,

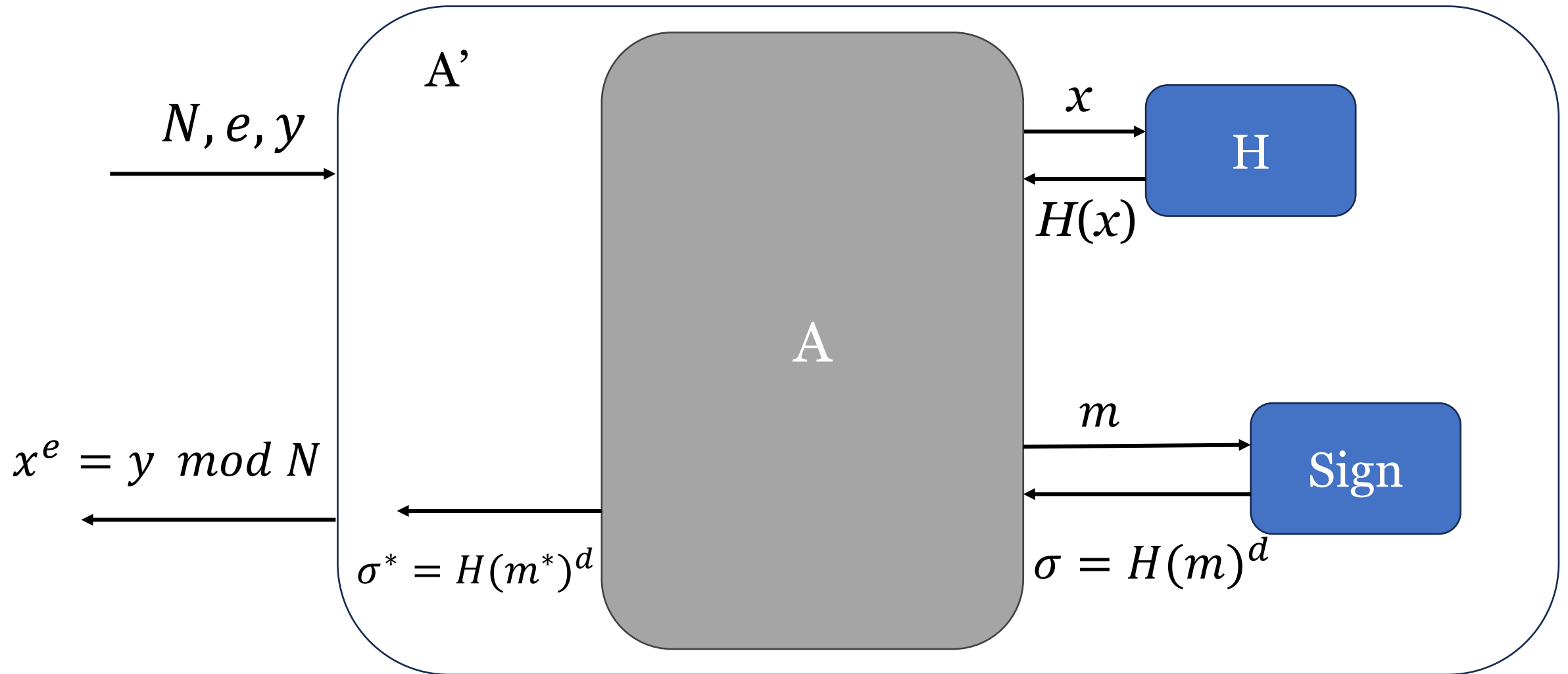
$$\Pr[\text{Sig_forge}_{A,\Pi}(n) = 1] \leq \text{negl}(n)$$

RSA assumption

1. Run $\text{Gen}(1^n)$ to obtain (N, e, d)
2. Choose a random $y \in \mathbb{Z}_N^*$
3. A is given (N, e, y) and outputs $x \in \mathbb{Z}_N^*$
4. Output 1 if $x^e = y \bmod N$

$$\Pr[\text{RSA}_A(n) = 1] \leq \text{negl}(n)$$

Security proof



Step 1

- Q : set of all queries to the signing oracle
- M : set of all queries to the random oracle H
 - Occurred at oracle to $\text{sign}(\cdot)$ and $\text{verify}(\cdot)$.

Claim 1: $m^* \notin Q, m^* \in M$, A must have queried m^* to H .

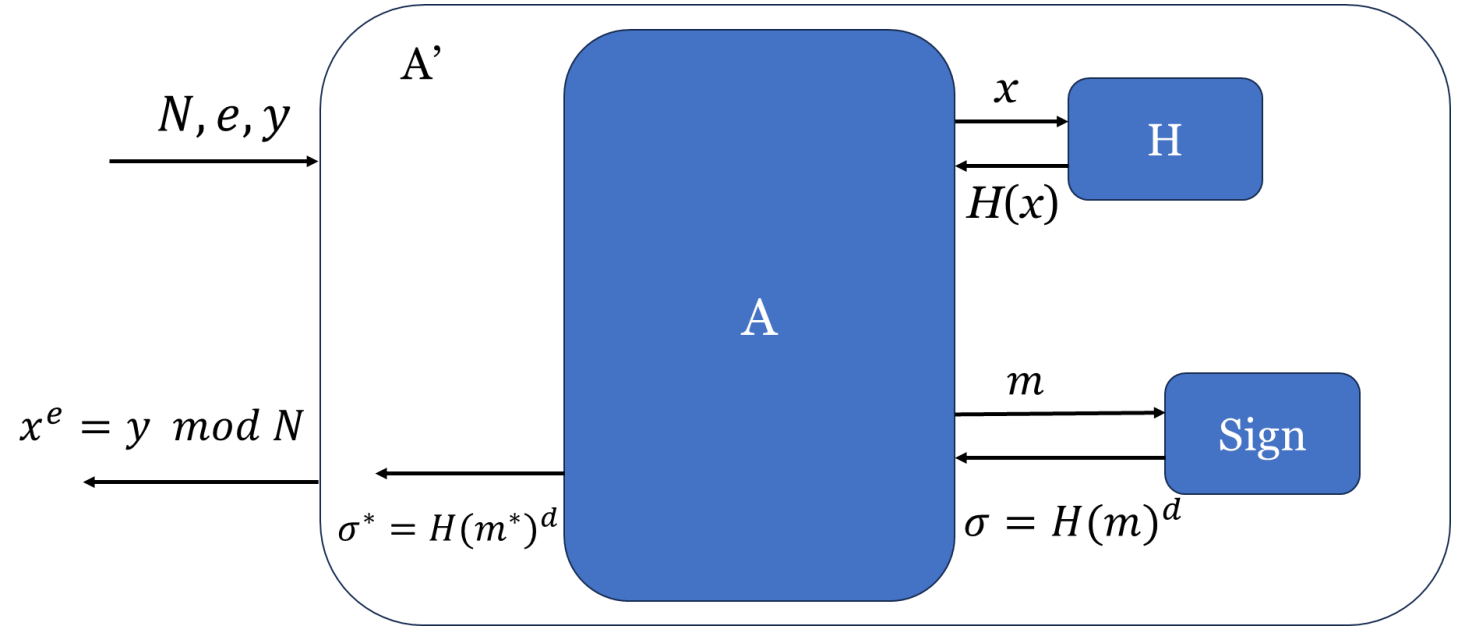
If not, $\sigma^{*e} = H(m^*)$ with negligible probability.

Suppose A makes $q(n)$ queries to H , $M = \{m_1, m_2, \dots, m_q\}$, $m^* = m_i$ for some $i \in [1, q]$

Step 2

- IF we know index i

1. Set $H(m_i) = y$
2. Obtain σ^* from A
3. $\sigma^{*e} = y$, output $x = \sigma^*$



RSA assumption

1. Run $\text{Gen}(1^n)$ to obtain (N, e, d)
2. Choose a random $y \in \mathbb{Z}_N^*$
3. A is given (N, e, y) and outputs $x \in \mathbb{Z}_N^*$
4. Output 1 if $x^e = y \bmod N$

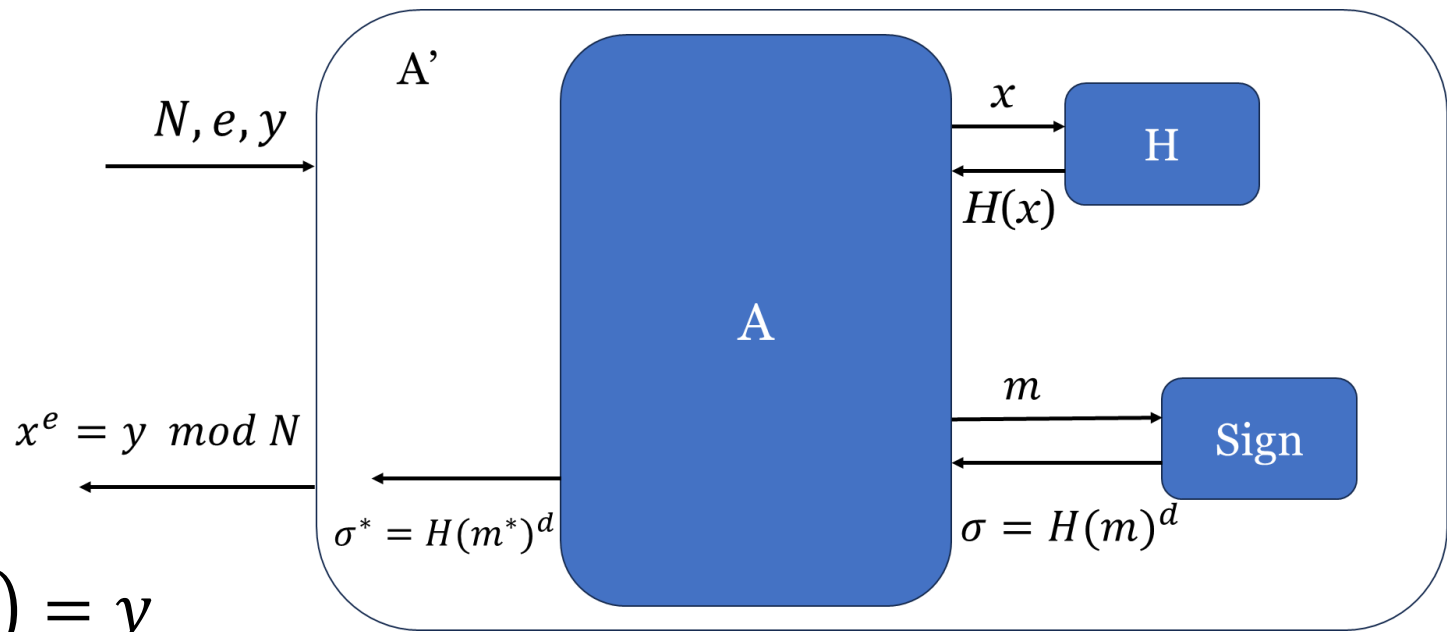
$$\Pr[\text{RSA}_A(n) = 1] \leq \text{negl}(n)$$

Step 3

Take a random guess: set $H(m_j) = y$

$$\Pr[i = j] = \frac{1}{q}$$

1. Pick a random $j \in [1, q]$
2. For $i = j$, set $H(m_i) = y$
3. For $i \neq j$, pick a random $\sigma_i \in \mathbb{Z}_N^*$
set $H(m_i) = \sigma_i^e \bmod N$



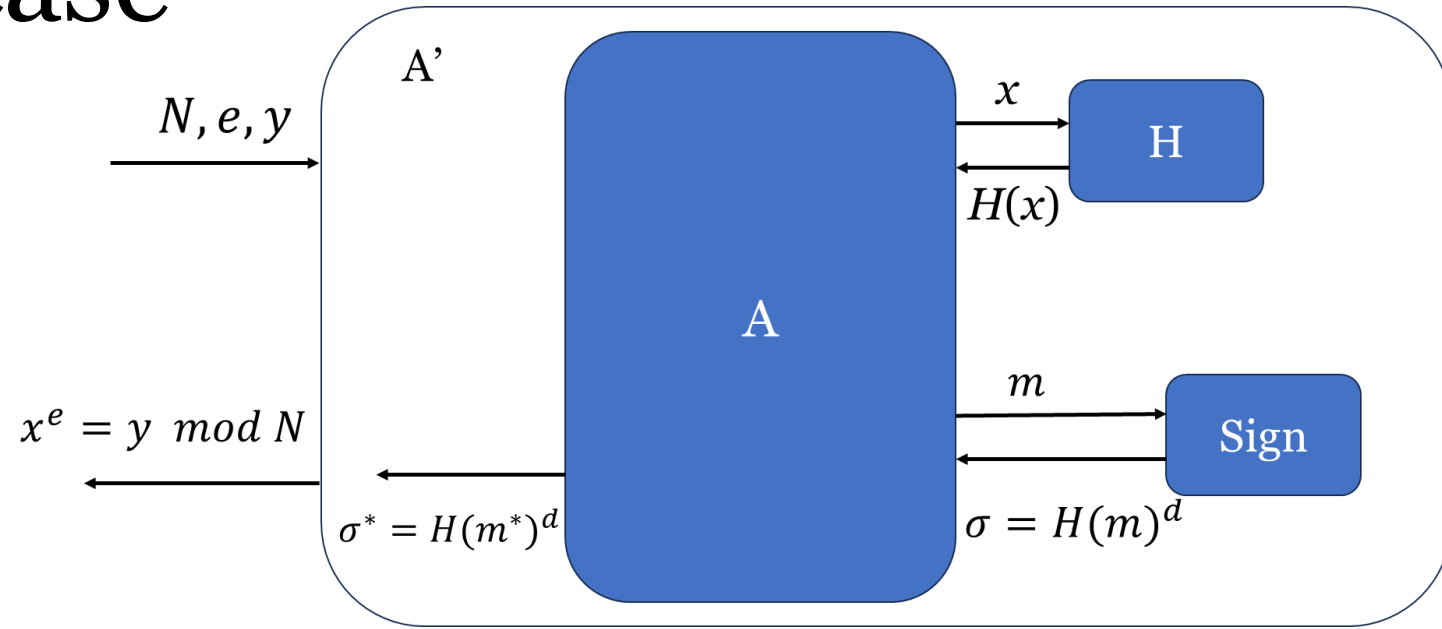
One more special case

Pick a random $j \in [1, q]$

set $H(m_j) = y$

What if A queries m_j to Sign?

- If it happens, abort.
- **Because m_j can't be the output forged message.**



Putting everything together

$A'(N, e, y)$

1. Choose uniform $j \in [1, q]$
2. Run A on input the public key $pk = (N, e)$
3. When A makes its i -th random-oracle query
 - If $i = j$, return $H(m_i) = y$
 - If $i \neq j$, pick a random $\sigma_i \in \mathbb{Z}_N^*$, return $H(m_i) = \sigma_i^e \bmod N$
4. When A makes a query to the signing oracle
 - If $m = m_j$, abort
 - If $m = m_i \neq m_j$, return σ_i
5. At the end, A outputs m^*, σ^* . If $m^* = m_j$ and $\sigma^{*e} = y$, output σ^*

Putting everything together

- $\Pr[\text{RSAInv}_{A', \text{GenRSA}}(n) = 1] = \Pr[A' \text{ succeeds}] = \frac{1}{q} \Pr[\text{Sig_forge}_{A, \Pi}(n) = 1]$
- Since q is polynomial, $\Pr[\text{RSAInv}_{A', \text{GenRSA}}(n) = 1]$ is negligible,
- $\Pr[\text{Sig_forge}_{A, \Pi}(n) = 1] = q \cdot \text{negl}(n)$ is also negligible.

Group, Ring and Field

Chinese Remainder Theorem

Group

Group: under 1 operation $\circ$

1. **Closure**: For all a, b in G , the result of the operation, $a \circ b$, is also in G
2. **Associativity**: For all a, b and c in G , $(a \circ b) \circ c = a \circ (b \circ c)$
3. **Identity element**: There exists an element e in G such that, for every element a in G , the equation $e \circ a = a \circ e = a$ holds. Such an element is unique
4. **Inverse element**: For each a in G , there exists an element b in G , such that $a \circ b = b \circ a = e$, where e is the identity element

Commutativity: If for all a, b in G , $a \circ b = b \circ a$, the group is called **abelian**

Examples: $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$ with $+$ mod N

(Finite) Field

2 operations $+$ and $\times$

- $\mathbb{F}$ is an **abelian** group under operation $+$
- $\mathbb{F} - \{0\}$ (the set without the additive identity 0) is an **abelian** group under operation $\times$.
- **(Distributivity:)** For all $a, b, c \in \mathbb{F}$, $a \cdot (b + c) = ab + ac$.

Examples:

- Integers with $+$ and $\times$ **No**
- Rational numbers with $+$ and $\times$ **Yes**
- Real numbers, complex numbers **Yes**
- $\mathbb{F}_p = \{0, 1, 2, \dots, p - 1\}$ **Yes, finite field**
- $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$ **No for composite N**

Ring

2 operations + and $\times$

- R is an **abelian** group under operation +
- **Associativity of $\times$** : For all a, b and c in R , $(a \times b) \times c = a \times (b \times c)$
- Distributive properties: For all a, b and c in R ,

$$a \times (b + c) = (a \times b) + (a \times c)$$

$$(b + c) \times a = (b \times a) + (c \times a)$$

Examples:

- Integers with + and $\times$
- Integers with + and $\times \bmod 2^{32}$

Every field is a ring and a group, and every ring is a group

Isomorphism

Definition 9.23: $\mathbb{G}, \mathbb{H}$ groups with operations $\circ_G, \circ_H$.

$f: \mathbb{G} \rightarrow \mathbb{H}$ is an isomorphism, denoted by $\mathbb{G} \cong \mathbb{H}$ if

1. f is a bijection
2. For all $a, b \in \mathbb{G}$, $f(a \circ_G b) = f(a) \circ_H f(b)$

Chinese Remainder Theorem

There are certain things whose number is unknown. If we count them by **threes**, we have two left over; by **fives**, we have three left over; and by **sevens**, two are left over. How many things are there?

-- Sunzi, 5th century

Chinese Remainder Theorem

- $x = 2 \bmod 3, \quad x = 3 \bmod 5, \quad x = 2 \bmod 7$
- Unique $x = 23$ for $x < 105 = 3 \times 5 \times 7$

Chinese Remainder Theorem

- $x \bmod N$
 - $(a_1 = x \bmod n_1, a_2 = x \bmod n_2, \dots, a_k = x \bmod n_k)$
- where $N = n_1 \times n_2 \times \dots \times n_k$ and n_i s are pairwise coprime.

We can define a **one-to-one** mapping from $\mathbb{Z}_N$ to $\mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$.

$$f(x) = (x \bmod n_1, \dots, x \bmod n_k), \text{ for each } x \in \mathbb{Z}_N.$$

Why one-to-one?

1. $\mathbb{Z}_N$ and $\mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$ has the same size.
2. $x_1 \neq x_2 \in \mathbb{Z}_N$ can't be mapped to the same $y \in \mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$

Chinese Remainder Theorem

$$\mathbb{Z}_N \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k}, \quad \mathbb{Z}_N^* \cong \mathbb{Z}_{n_1}^* \times \mathbb{Z}_{n_2}^* \times \cdots \times \mathbb{Z}_{n_k}^*$$

under the one-to-one mapping $f(\cdot)$, $n_1, \dots, n_k$ coprime.

• **Operations:** $(a_1, \dots, a_k) + (b_1, \dots, b_k) := (a_1 + b_1, \dots, a_k + b_k)$, $\times$: similarly.

Example: $N = 15 = 5 \times 3$

$0 \leftrightarrow (0, 0)$	$1 \leftrightarrow (1, 1)$	$2 \leftrightarrow (2, 2)$	$3 \leftrightarrow (3, 0)$
$4 \leftrightarrow (4, 1)$	$5 \leftrightarrow (0, 2)$	$6 \leftrightarrow (1, 0)$	$7 \leftrightarrow (2, 1)$
$8 \leftrightarrow (3, 2)$	$9 \leftrightarrow (4, 0)$	$10 \leftrightarrow (0, 1)$	$11 \leftrightarrow (1, 2)$
$12 \leftrightarrow (2, 0)$	$13 \leftrightarrow (3, 1)$	$14 \leftrightarrow (4, 2)$	

What is $13 \times 14 \bmod 15$?

$$13(3,1) \times 14(4,2) = 2(2,2)$$

What is $13 + 14 \bmod 15$?

$$13(3,1) + 14(4,2) = 12(2,0)$$

How to compute $f^{-1}(\cdot)$?

$$f(x) = (x \bmod n_1, \dots, x \bmod n_k)$$

But given $(x_1, x_2, \dots, x_k)$, how to compute $x \in \{0, \dots, N - 1\}$?

My favorite method:

1. Construct

$$(1, 0, \dots, 0) \rightarrow B_1 \in \mathbb{Z}_N$$

$$(0, 1, \dots, 0) \rightarrow B_2 \in \mathbb{Z}_N$$

$$\vdots$$

$$(0, \dots, 0, 1) \rightarrow B_k \in \mathbb{Z}_N$$

2. Compute $x = (x_1 B_1 + x_2 B_2 + \dots + x_k B_k) \bmod N$.